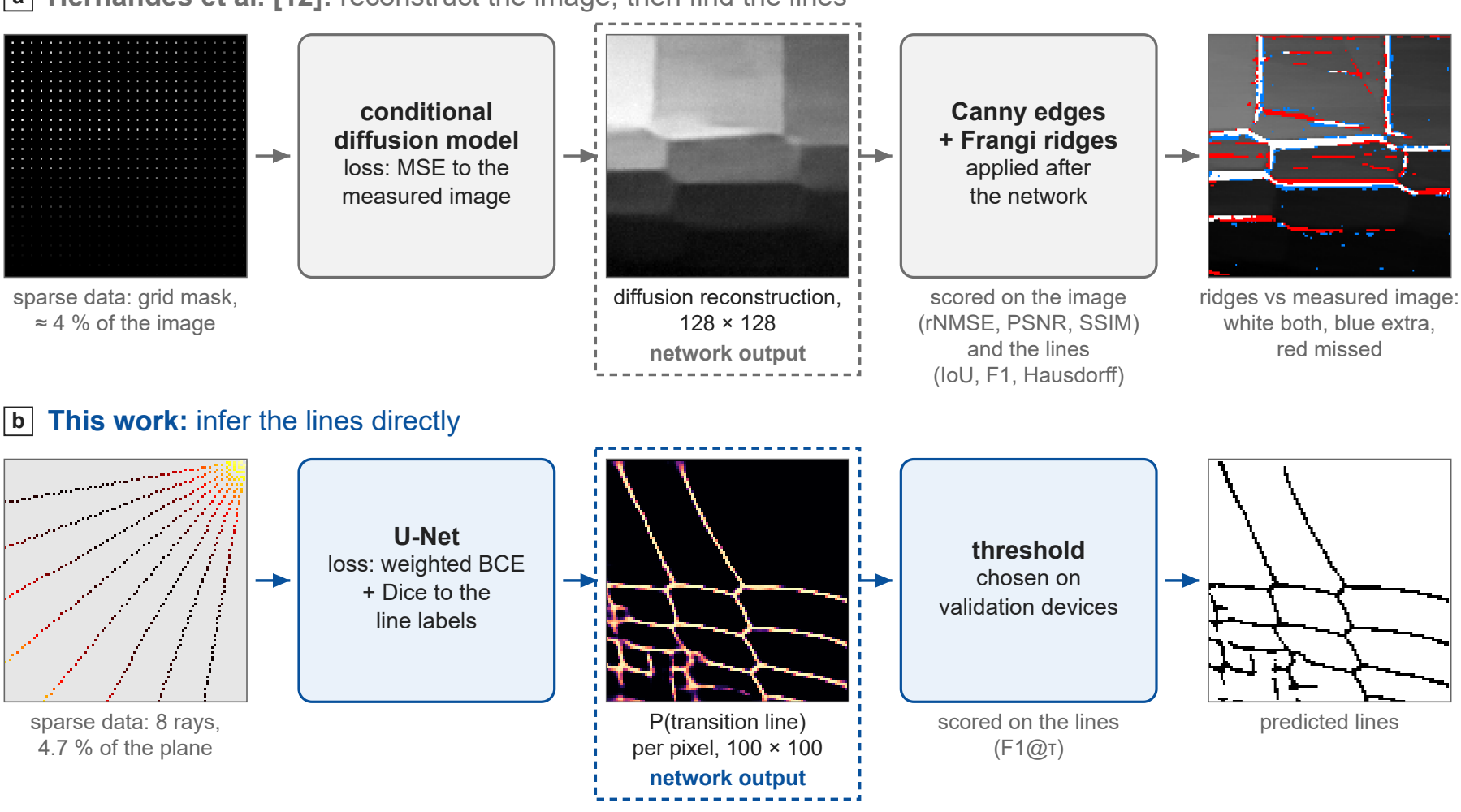

a Hernandes et al. [12]: reconstruct the image, then find the lines
conditional diffusion model
loss: MSE to the measured image
Canny edges + Frangi ridges
applied after the network
sparse data: grid mask, ≈ 4 % of the image
diffusion reconstruction, 128 × 128
network output
scored on the image (rNMSE, PSNR, SSIM) and the lines (IoU, F1, Hausdorff)
ridges vs measured image: white both, blue extra, red missed
b This work: infer the lines directly
U-Net
loss: weighted BCE + Dice to the line labels
threshold
chosen on validation devices
sparse data: 8 rays, 4.7 % of the plane
P(transition line) per pixel, 100 × 100
network output
scored on the lines (F1@τ)
predicted lines